SGN – Assignment #2

Davide Bellini, 247658

Exercise 1: Uncertainty propagation

You are asked to analyze the state uncertainty evolution along a transfer trajectory in the Planar Bicircular Restricted Four-Body Problem, obtained as optimal solution of the problem stated in Section 3.1 (Topputo, 2013)*. The mean initial state $\mathbf{x}_i$ at initial time t_i with its associated covariance $\mathbf{P}_0$ and final time t_f for this optimal transfer are provided in Table 1.

1. Propagate the initial mean and covariance within a time grid of 5 equally spaced elements going from t_i to t_f, using both a Linearized Approach (LinCov) and the Unscented Transform (UT). We suggest to use $\alpha = 1$ and $\beta = 2$ for tuning the UT in this case. Plot the mean and the ellipses associated with the position elements of the covariances obtained with the two methods at the final time.

2. Perform the same uncertainty propagation process on the same time grid using a Monte Carlo (MC) simulation †. Compute the sample mean and sample covariance and compare them with the estimates obtained at Point 1). Provide the following outputs.

   - Plot of the propagated samples of the MC simulation, together with the mean and the covariance obtained with all methods in terms of ellipses associated with the position elements at the final time.
   - Plot of the time evolution (for the time grid previously defined) for all three approaches (MC, LinCov, and UT) of $3\sqrt{\max(\lambda_i(P_r))}$ and $3\sqrt{\max(\lambda_i(P_v))}$, where P_r and P_v are the 2x2 position and velocity covariance submatrices.
   - Plot resulting from the use of the MATLAB function `qqplot`, for each component of the previously generated MC samples at the final time.

   Compare the results, in terms of accuracy and precision, and discuss on the validity of the linear and Gaussian assumption for uncertainty propagation.

Table 1: Solution for an Earth-Moon transfer in the rotating frame.

Parameter	Value
Initial state $\mathbf{x}_i$	$\mathbf{r}_i$ = [-0.011965533749906, -0.017025663128129] $\mathbf{v}_i$ = [10.718855256727338, 0.116502348513671]
Initial time t_i	1.282800225339865
Final time t_f	9.595124551366348
Covariance $\mathbf{P}_0$	$\begin{bmatrix} +1.041e-15 & +6.026e-17 & +5.647e-16 & +4.577e-15 \\ +6.026e-17 & +4.287e-18 & +4.312e-17 & +1.855e-16 \\ +5.647e-16 & +4.312e-17 & +4.432e-16 & +1.455e-15 \\ +4.577e-15 & +1.855e-16 & +1.455e-15 & +2.822e-14 \end{bmatrix}$

*F. Topputo, "On optimal two-impulse Earth–Moon transfers in a four-body model", Celestial Mechanics and Dynamical Astronomy, Vol. 117, pp. 279–313, 2013, DOI: 10.1007/s10569-013-9513-8.

†Use at least 1000 samples drawn from the initial covariance

Exercise 1: Uncertainty Propagation

1.1 Mean and Covariance Propagation

Methodology The mean and covariance of the spacecraft state are propagated over the time interval using two different methods: the Linearized Covariance (LinCov) and the Unscented Transform (UT). A time grid of five equally spaced points is used between the initial time t_0 and the final time t_f. The initial state vector $\mathbf{x}_0$ and covariance matrix $\mathbf{P}_0$, as defined in the assignment, served as inputs for both methods.

In the **Linearized Covariance** , it is assumed that uncertainties follow a normal distribution and that the dynamics can be approximated linearly around the mean state using the State Transition Matrix (STM). The mean and covariance at each time step t_i are propagated using the following equations:

$$\hat{\mathbf{x}}_{LinCov}(t_i) = \varphi(t_0, t_i; \mathbf{x}_0) \qquad \mathbf{P}_{LinCov}(t_i) = \mathbf{\Phi}(t_0, t_i)\mathbf{P}_0\mathbf{\Phi}^T(t_0, t_i) \tag{1}$$

where $\hat{\mathbf{x}}_{LinCov}(i)$ and $\mathbf{P}_{LinCov}(i)$ are the estimated mean and covariance at time t_i, respectively, φ is the flow and Φ is the STM.

For the **Unscented Transform**, a set of sigma points $\boldsymbol{\chi_j}$ is generated around the initial mean to parametrize the initial Probability Density Function (PDF). These sigma points are then propagated with the Planar Bi-circular Restricted Four-Body Problem (PBR4BP) at each time step such that $\gamma_j = f(\chi_j)$. The propagated sigma points $\boldsymbol{\gamma_j}$ are used to compute the mean and covariance with weighted averages through the matrices W^m, W^c :

$$\hat{\mathbf{x}}_{UT}(t_i) = \sum_{j=0}^{2n} W_j^m \boldsymbol{\gamma_j} \qquad \mathbf{P}_{UT}(t_i) = \sum_{j=0}^{2n} W_j^c (\boldsymbol{\gamma_j} - \hat{\mathbf{x}}_{UT}(t_i))(\boldsymbol{\gamma_j} - \hat{\mathbf{x}}_{UT}(t_i))^T \tag{2}$$

The final results are visualized by plotting the mean positions and 3σ confidence ellipses for both methods at t_f. The ellipses are constructed from the eigenvectors and eigenvalues of the position covariance submatrix.

Results The results indicate that the *UT* method gives a more precise estimation of the mean and covariance compared to *LinCov*. This difference may arise from the initial assumptions of on linear dynamics approximation and Gaussian error distribution made by *LinCov* , which fail to fully account for the non-linearities of the PBR4BP. In contrast, the *UT* assumes normality only for the initial distribution, enabling a more accurate uncertainty propagation through the non-linear dynamics. As shown in Fig. 1, The distorted covariance ellipse observed in *LinCov* suggests an underestimation of uncertainties, especially along secondary trajectory directions, while the *UT* method more effectively captures the error distribution. However, the close agreement in the mean states between the two methods implies that the system's dynamics exhibit limited non-linearity within the considered propagation period.

1.2 Monte Carlo Simulation

Methodology A Monte Carlo (MC) simulation is performed with random population of 1000 samples generated using the initial mean $\mathbf{x}_0$ and covariance $\mathbf{P}_0$ with the MATLAB function `mvnrnd`. At each time step t_i, the samples are propagated with the PBR4BP, and the sample mean $\hat{\mathbf{x}}_{MC}(t_i)$ and sample covariance $\mathbf{P}_{MC}(t_i)$ are calculated from:

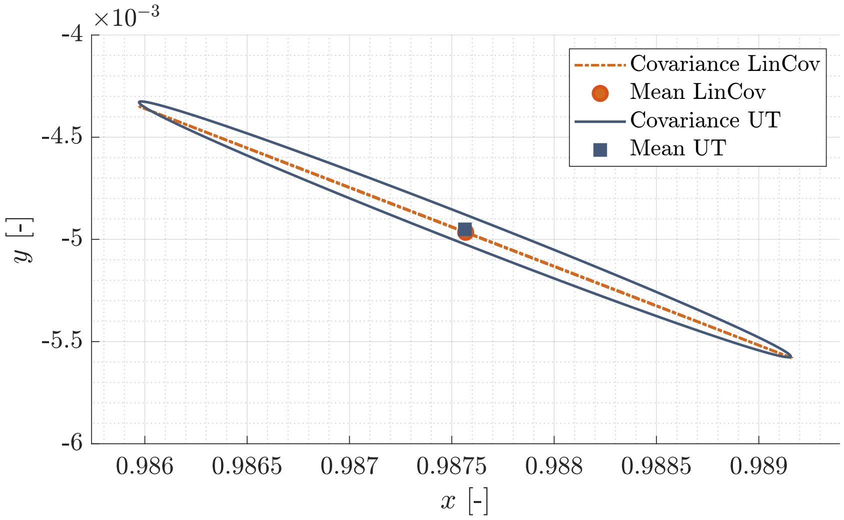


Figure 1: Mean positions and covariance ellipses at final time t_f for both *LinCov* and *UT*.

$$\hat{\mathbf{x}}_{MC}(t_i) = \frac{1}{N}\sum_{j=1}^{N}\mathbf{y}_j \qquad \mathbf{P}_{MC}(t_i) = \frac{1}{N-1}\sum_{j=1}^{N}(\mathbf{y}_j - \hat{\mathbf{x}}_{MC}(t_i))(\mathbf{y}_j - \hat{\mathbf{x}}_{MC}(t_i))^T \tag{3}$$

where N is the population size and $\mathbf{y}_j$ are the propagated samples at each time step.

As described in Section 1.1, the confidence ellipses at the final time t_f are computed and compared with the results obtained from the previous two methods. Subsequently, for all three approaches, the maximum eigenvalues of the position ($\mathbf{P}_r$) and velocity ($\mathbf{P}_v$) covariance submatrices are calculated, and the values of $3\sqrt{\max(\boldsymbol{\lambda}_i(\mathbf{P}_r))}$ and $3\sqrt{\max(\boldsymbol{\lambda}_i(\mathbf{P}_v))}$ are plotted and compared. Finally, the QQ-plots are generated for each state component at the final time to assess the Gaussianity of the propagated samples.

Results From Fig. 2 it is possible to observe that the UT method closely matches the MC results, indicating again that it captures non-linear effects more accurately than *LinCov*. This comparison highlights that *LinCov* deviates from the MC benchmark, while *UT* provides a more robust representation of uncertainty in non-linear dynamics.

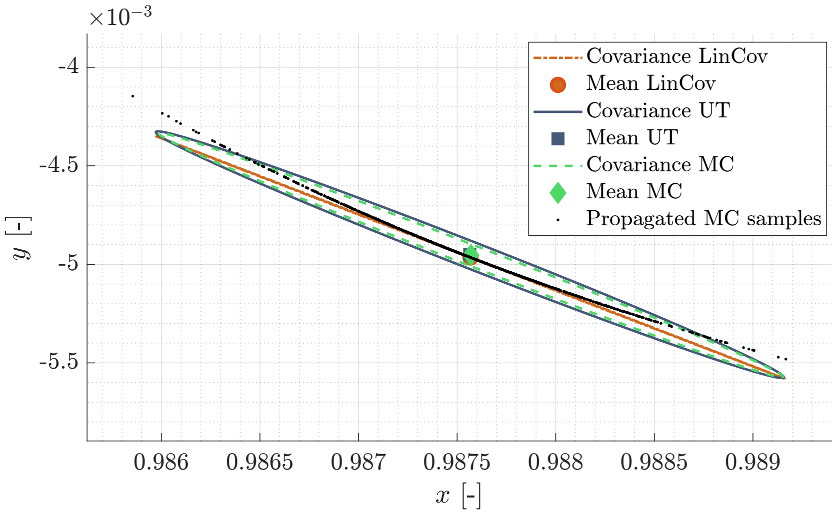
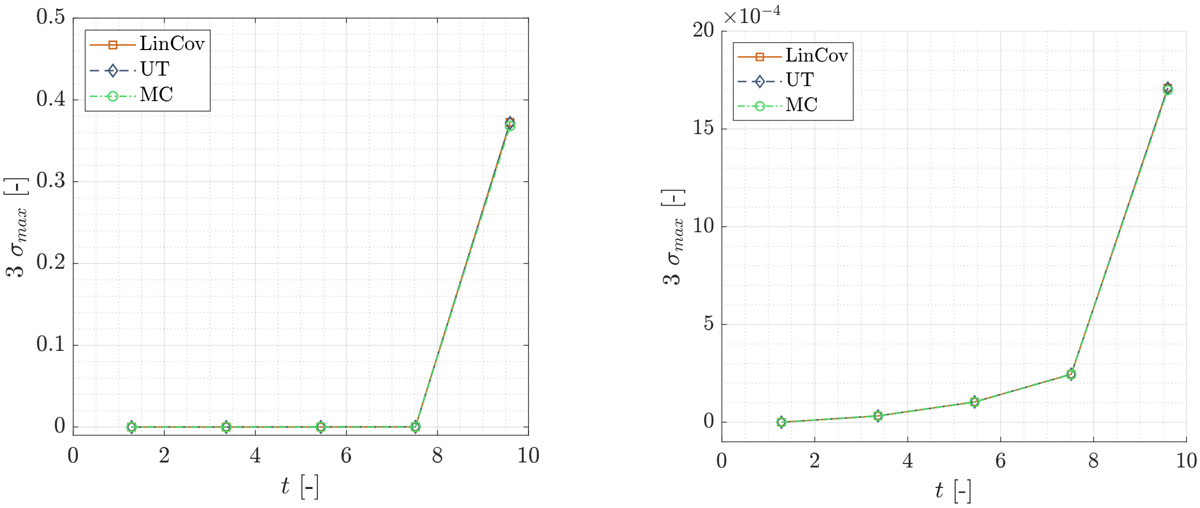


Figure 2: Mean positions and covariance ellipses at final time t_f for LinCov, UT, and MC.

The time evolution of the maximum standard deviation $3\sigma_{\max}$ for both position and velocity covariance matrices, presented in Fig. 3 confirm that *LinCov* performs comparably to the other methods in the primary propagation direction. However, MC and UT show superior accuracy overall.



(a) Position. **(b)** Velocity.

Figure 3: Time evolution of $3\sqrt{\max(\lambda(\mathbf{P}_r))}$ and $3\sqrt{\max(\lambda(\mathbf{P}_v))}$ for LinCov, UT, and MC.

From the QQ-plots in Fig. 4, it is possible to observe that the Gaussianity assumption is valid overall. The only deviation arises in the $x-$component of the velocity v_x where the points are not well aligned with the reference path.

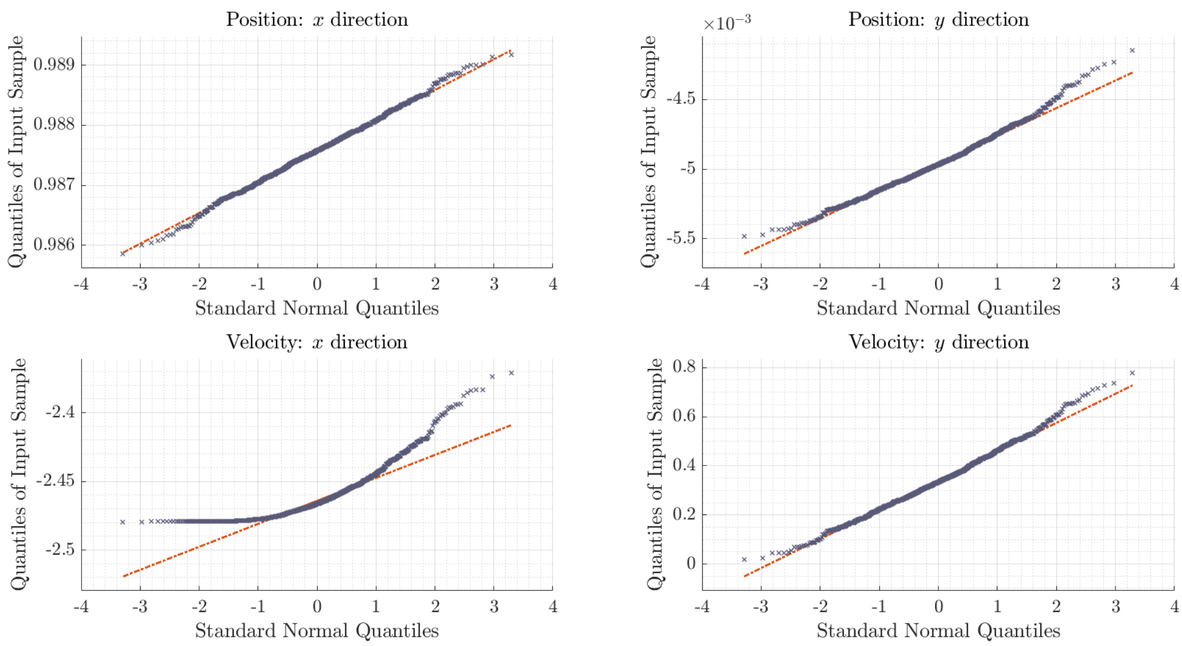


Figure 4: QQ-plots of position and velocity components for MC samples at t_f.

From the performed analysis, it is found that the MC simulation provided the most accurate results but present the drawback of a high computational cost. The *LinCov* is the more computationally efficient, but provides less reliable solutions when non-linearities are present. The UT method balanced accuracy and efficiency, making it the optimal approach for non-linear regimes.

Exercise 2: Batch filters

The Soil Moisture and Ocean Salinity (SMOS) mission, launched on 2 November 2009, is one of the European Space Agency's Earth Explorer missions, which form the science and research element of the Living Planet Programme.

You have been asked to track SMOS to improve the accuracy of its state estimate. To this aim, you shall schedule the observations from the three ground stations reported in Table 2.

1. *Compute visibility windows.* The Two-Line Elements (TLE) set of SMOS are reported in Table 3 (and in WeBeep as 36036.3le). Compute the osculating state from the TLE at the reference epoch t_{ref}, then propagate this state assuming Keplerian motion to predict the trajectory of the satellite and compute all the visibility time windows from the available stations in the time interval from $t_0 =$ `2024-11-18T20:30:00.000` (UTC) to $t_f =$ `2024-11-18T22:15:00.000` (UTC). Consider the different time grid for each station depending on the frequency of measurement acquisition. Report the resulting visibility windows and plot the predicted Azimuth and Elevation profiles within these time intervals.

2. *Simulate measurements.* Use SGP4 and the provided TLE to simulate the measurements acquired by the sensor network in Table 2 by:

   (a) Computing the spacecraft position over the visibility windows identified in Point 1 and deriving the associated expected measurements.

   (b) Simulating the measurements by adding a random error to the expected measurements (assume a Gaussian model to generate the random error, with noise provided in Table 2). Discard any measurements (i.e., after applying the noise) that does not fulfill the visibility condition for the considered station.

3. *Solve the navigation problem.* Using the measurements simulated at the previous point:

   (a) Find the least squares (minimum variance) solution to the navigation problem without a priori information using
      - the epoch t_0 as reference epoch;
      - the reference state as the state derived from the TLE set in Table 3 at the reference epoch;
      - the simulated measurements obtained for the KOROU ground station only;
      - pure Keplerian motion to model the spacecraft dynamics.

   (b) Repeat step 3a by using all simulated measurements from the three ground stations.

   (c) Repeat step 3b by using a J2-perturbed motion to model the spacecraft dynamics.

   Provide the results in terms of navigation solution[‡], square root of the trace of the estimated covariance submatrix of the position elements, square root of the trace of the estimated covariance submatrix of the velocity elements. Finally, considering a linear mapping of the estimated covariance from Cartesian state to Keplerian elements, provide the standard deviation associated to the semimajor axis, and the standard deviation associated to the inclination. Elaborate on the results, comparing the different solutions.

4. *Trade-off analysis.* For specific mission requirements, you are constrained to get a navigation solution within the time interval reported in Point 1. Since the allocation of antenna time has a cost, you are asked to select the passes relying on a budget of 70.000 €. The cost per pass of each ground station is reported in Table 2. Considering this constraint,

[‡]Not just estimated state or covariance

and by using a J2-perturbed motion for your estimation operations, select the best combination of ground stations and passes to track SMOS in terms of resulting standard deviation on semimajor axis and inclination, and elaborate on the results.

5. *Long-term analysis.* Consider a nominal operations scenario (i.e., you are not constrained to provide a navigation solution within a limited amount of time). In this context, or for long-term planning in general, you could still acquire measurements from multiple locations but you are tasked to select a set of prime and backup ground stations. For planning purposes, it is important to have regular passes as this simplifies passes scheduling activities. Considering the need to have *reliable* orbit determination and *repeatable* passes, discuss your choices and compare them with the results of Point 4.

Table 2: Sensor network to track SMOS: list of stations, including their features.

Station name	KOUROU	TROLL	SVALBARD
Coordinates	LAT = 5.25144° LON = -52.80466° ALT = -14.67 m	LAT = -72.011977° LON = 2.536103° ALT = 1298 m	LAT = 78.229772° LON = 15.407786° ALT = 458 m
Type	Radar (monostatic)	Radar (monostatic)	Radar (monostatic)
Measurements type	Az, El [deg] Range (one-way) [km]	Az, El [deg] Range (one-way) [km]	Az, El [deg] Range (one-way) [km]
Measurements noise (diagonal noise matrix R)	$\sigma_{Az,El} = 125$ mdeg $\sigma_{range} = 0.01$ km	$\sigma_{Az,El} = 125$ mdeg $\sigma_{range} = 0.01$ km	$\sigma_{Az,El} = 125$ mdeg $\sigma_{range} = 0.01$ km
Minimum elevation	6 deg	0 deg	8 deg
Measurement frequency	60 s	30 s	60 s
Cost per pass	30.000 €	35.000 €	35.000 €

Table 3: TLE of SMOS.

```
1␣36036U␣09059A␣␣␣24323.76060260␣␣.00000600␣␣00000-0␣␣20543-3␣0␣␣9995
2␣36036␣␣98.4396␣148.4689␣0001262␣␣95.1025␣265.0307␣14.39727995790658
```

Exercise 2: Batch Filters

2.1 Visibility Windows

Methodology The initial data for the SMOS satellite was extracted from the Two-Line Elements (TLE) provided in the assignment. Through SGP4 the SMSOS state at the reference epoch t_{ref} is computed. Subsequently, time values are converted first to Julian Day and then into elapsed time from the J2000 epoch. To perform the propagation, the spacecraft state $\mathbf{x}_{ref}$ is rotated from the True Equator Mean Equinox (TEME) reference frame to the Earth-Centered Inertial (ECI) reference frame, the state is then propagated with Keplerian motion from the reference epoch t_{ref} to the mission start time t_0, and further to the final time t_f.

To determine the visibility windows relative to the three available ground stations, the azimuth, range, and elevation values are computed. For these computations, the position and velocity of each ground station are extracted from the provided kernel over the interval from t_0 to t_f. The difference between the station's state and the previously propagated state of SMOS was then calculated to obtain the relative state. This relative state was transformed into the Topocentric (TOPO) reference frame of the ground station. Using the SPICE function `cspice_xfmsta`, the azimuth, range, and elevation values were accurately derived. This process is repeated independently for all three ground stations by varying the measurement acquisition frequency based on the capabilities of each GS (Kourou: 60 s, Troll: 30 s, Svalbard: 60 s).

The visibility windows were identified by selecting the time intervals during which the computed elevation exceeds the station-specific minimum elevation thresholds. Within these visibility windows, the time evolution of the satellite's azimuth and elevation relative to the ground stations was analyzed.

Results The osculating state, derived from the TLE is presented in Table 4. The visibility windows for the three ground stations (GS) are summarized in Table 5, indicating the time intervals during which the satellite is within the observable range.

x [km]	y [km]	z [km]	v_x [km/s]	v_y [km/s]	v_z [km/s]
-6065.4138	3768.0469	14.5010	0.603625	0.926743	7.390768

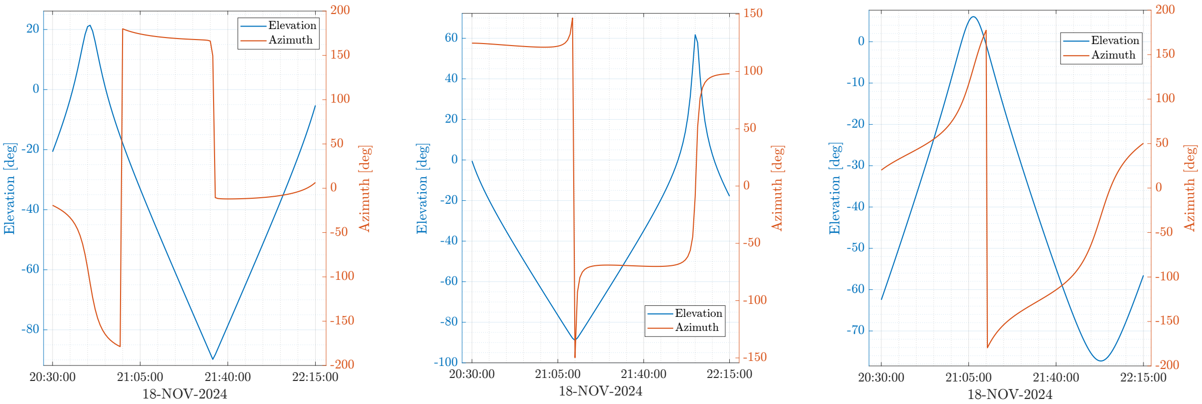
Table 4: Osculating state at the reference epoch.

Ground Station	Start Time	End Time
Kourou	2024 NOV 18 20:40:00.000	2024 NOV 18 20:49:00.000
Troll	2024 NOV 18 21:02:30.000	2024 NOV 18 21:11:30.000
Svalbard	2024 NOV 18 21:56:00.000	2024 NOV 18 22:06:00.000

Table 5: Visibility windows for the three ground stations.

Fig. 5 shows the time evolution of azimuth and elevation, for each ground station, during the respective observation intervals.

Additionally, in Fig. 6, the elevation vs. azimuth profiles for the three ground stations are shown within their corresponding visibility windows, highlighting the satellite's trajectory as observed from each GS.



(a) Kourou **(b)** Troll **(c)** Svalbard

Figure 5: Time evolution of azimuth and elevation for the three ground stations.

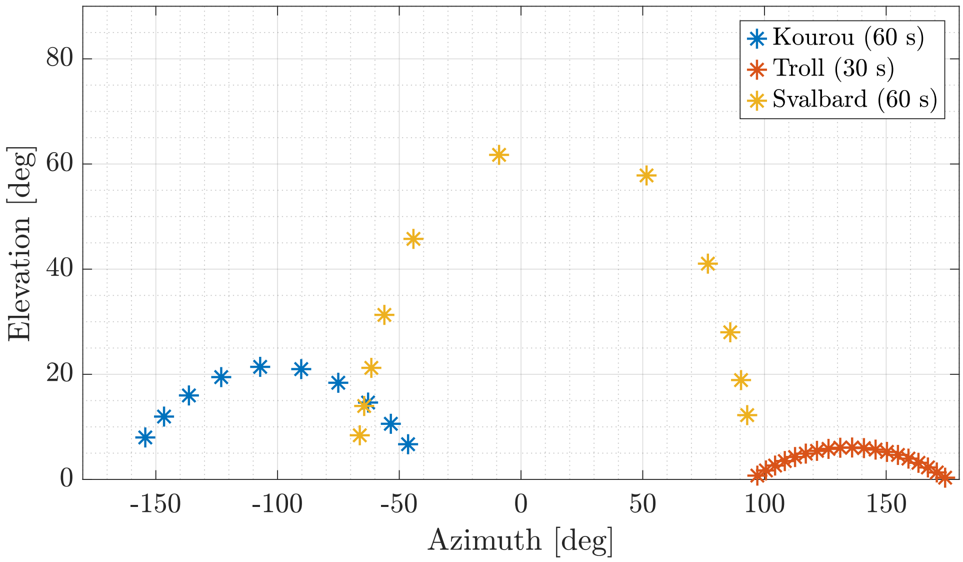


Figure 6: Elevation vs. azimuth profiles for the three ground stations during their respective visibility windows.

2.2 Simulated Measurements

Methodology The SGP4 propagator is employed to compute the reference state of the SMOS satellite within the visibility windows identified in Section 2.1, for each ground station. Using the procedure described in Section 2.1, the azimuth, elevation, and range values are calculated.

To simulate realistic measurements, random noise was added to the computed values of azimuth, elevation, and range. The noise was modelled as Gaussian with zero mean and covariance σ^2, as specified in GS specifications. The MATLAB function `mvnrnd` was used to generate the perturbations. The perturbed measurements are then compared with the station-specific visibility conditions. Any measurements outside the visibility constraints, due to perturbation, are discarded.
Finally, the azimuth and elevation values, both with and without noise, are plotted to analyze the difference in the satellite's trajectory visibility windows using different propagation models.

Results Fig. 7 shows that the Keplerian and SGP4 models generate azimuth and elevation measurements with some discrepancies, as expected due to their differing levels of dynamical accuracy.

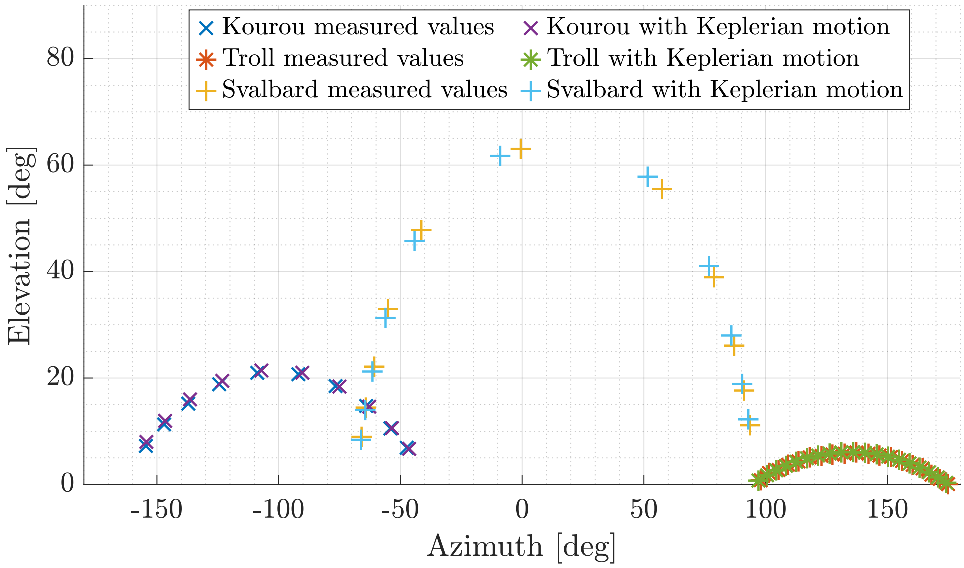


Figure 7: Measured values of elevation and azimuth for the three ground stations

2.3 Solution to the Navigation Problem

Methodology In this section, the navigation problem is addressed using a batch filter. The spacecraft dynamics are modelled assuming Keplerian motion, and the problem is solved by minimizing the residuals between the azimuth, elevation, and range obtained from the propagation, as described in the previous section, and the corresponding simulated measurements. The residual minimization is performed using the MATLAB function `lsqnonlin`, and providing as input the cost function presented in Algorithm 1. The simulated measurements are taken from the results of Section 2.2, where SGP4 was used for propagation, including noise.

Algorithm 1 Cost Function Navigation Problem

Require: x_0: Initial state vector, t_{span}: Propagation time, $Stations$: Ground stations data, $ODEfun$: Spacecraft dynamics function
Ensure: $residual$: Residual vector between predicted and real measurements
  Initialize $n_{\text{Stations}} \leftarrow$ number of ground stations
  Propagate spacecraft state $\leftarrow$ `ode113`$(ODEfun, t_{\text{span}}, x_0)$
  Pre-allocate $residual$ vector with size based on total number of measurements
  for each ground station i **do**
    Extract station data:
      $t_{\text{vis}} \leftarrow$ visibility window
      $W_m \leftarrow$ measurement weights
      $\text{meas}_{\text{real}} \leftarrow$ real measurements
    Identify valid measurement times in t_{vis}
    Compute predicted measurements: $\text{meas}_{\text{pred}}$
    Calculate residuals:
      $residual_{station_i} \leftarrow W_m(\text{meas}_{\text{pred}} - \text{meas}_{\text{real}})$
    Append $residual_{station_i}$ to $residual$
  end for
    return $residual$

The problem is solved for three configurations. First, the measurements from the Kourou ground station only are used. Then, the complete set of measurements from all three ground stations is considered. Finally, including again all three ground stations, but implementing a more accurate propagation model that accounts for accelerations due to Earth's oblateness (J2).

After obtaining the navigation solution (mean and covariance) for each configuration, a linear mapping is performed to evaluate the uncertainties in the semimajor axis (SMA) and inclination (i). The Jacobian matrix, representing the transformation from Cartesian coordinates to Keplerian elements, is computed using the forward finite difference. The covariance in the Keplerian domain is calculated as:

$$\mathbf{P}_{\text{kep}} = \mathbf{J} \cdot \mathbf{P}_{\text{cart}} \cdot \mathbf{J}^T \tag{4}$$

From the resulting covariance matrix $\mathbf{P}_{\text{kep}}$, the standard deviations σ_{SMA} and σ_i are extracted.

Results The results for the Kourou ground station only are presented in Table 6. This scenario shows high uncertainties both in position and velocity estimation due to the limited measurements available. Clearly, being SMOS in a polar orbit, the satellite has limited visibility from Kourou due to its equatorial location and to the 6° minimum elevation requirement. From Table 7 it is clear that the inclusion of measurements from Troll and Svalbard significantly improves the accuracy of the navigation solution. The increased number of measurements available together with the better geographical ground stations coverage reduce the dependence on a single station and enhance the solution redundancy. Finally, Table 8 highlights the precision achieved by incorporating the J2-perturbed dynamics. This dynamic model produces measurements closer to the ones obtained with SGP4, resulting in a more accurate final estimation of the spacecraft's state.

Parameter	**Value**
Position in ECI $[x, y, z]$ [km]	$[3933.3822, -1420.6625, 5783.1612]$
Velocity in ECI $[v_x, v_y, v_z]$ [km/s]	$[4.879043, -3.759073, -4.240784]$
$\sigma(P_r)$ [km]	11.3783
$\sigma(P_v)$ [km/s]	0.009935
$\sigma(SMA)$ [km]	8.6621
$\sigma(i)$ [deg]	0.087476

$$\mathbf{P_0} = \begin{bmatrix} 47.9742 & 22.8535 & -17.2961 & -4.3270e-02 & -3.1214e-02 & -2.1048e-02 \\ 22.8535 & 45.9714 & -40.1398 & -5.9850e-03 & -3.6045e-02 & 8.3717e-03 \\ -17.2961 & -40.1398 & 35.5210 & 2.3802e-03 & 2.9903e-02 & -9.3960e-03 \\ -4.3270e-02 & -5.9850e-03 & 2.3802e-03 & 4.5153e-05 & 1.9137e-05 & 2.6583e-05 \\ -3.1214e-02 & -3.6045e-02 & 2.9903e-02 & 1.9137e-05 & 3.4440e-05 & 3.1388e-06 \\ -2.1048e-02 & 8.3717e-03 & -9.3960e-03 & 2.6583e-05 & 3.1388e-06 & 1.9117e-05 \end{bmatrix}$$

Table 6: Navigation Results - Kourou Measurements Only

2.4 Trade-Off Analysis

Methodology With a limited budget of €70,000, only two ground stations could be used. Therefore, the three possible combinations of stations Kourou-Troll, Kourou-Svalbard, and

Parameter	Value
Position in ECI $[x, y, z]$ [km]	$[3926.6052, -1410.6447, 5780.1157]$
Velocity in ECI $[v_x, v_y, v_z]$ [km/s]	$[4.888920, -3.765053, -4.233128]$
$\sigma(P_r)$ [km]	0.944417
$\sigma(P_v)$ [km/s]	0.000934
$\sigma(SMA)$ [km]	0.189297
$\sigma(i)$ [deg]	0.002113

$$\mathbf{P_0} = \begin{bmatrix} 0.4697 & -0.2964 & -0.0734 & -4.5270e-04 & 1.6675e-04 & -3.0288e-04 \\ -0.2964 & 0.3468 & 0.0804 & 3.3901e-04 & -8.4568e-05 & 2.5195e-04 \\ -0.0734 & 0.0804 & 0.0755 & 3.7455e-05 & -1.8905e-05 & 8.8473e-05 \\ -4.5270e-04 & 3.3901e-04 & 3.7455e-05 & 5.1922e-07 & -1.2661e-07 & 2.9748e-07 \\ 1.6675e-04 & -8.4568e-05 & -1.8905e-05 & -1.2661e-07 & 1.0525e-07 & -1.0456e-07 \\ -3.0288e-04 & 2.5195e-04 & 8.8473e-05 & 2.9748e-07 & -1.0456e-07 & 2.4791e-07 \end{bmatrix}$$

Table 7: Navigation Results - All Ground Stations

Parameter	Value
Position in ECI $[x, y, z]$ [km]	$[3932.7779, -1414.9228, 5778.4941]$
Velocity in ECI $[v_x, v_y, v_z]$ [km/s]	$[4.879795, -3.763208, -4.232886]$
$\sigma(P_r)$ [km]	0.035753
$\sigma(P_v)$ [km/s]	0.000039
$\sigma(SMA)$ [km]	0.002372
$\sigma(i)$ [deg]	0.000056

$$\mathbf{P_0} = \begin{bmatrix} 8.4157e-04 & -4.6398e-04 & -2.1998e-04 & -8.4608e-07 & 2.8907e-07 & -5.8198e-07 \\ -4.6398e-04 & 3.2442e-04 & 1.4754e-04 & 4.9912e-07 & -1.4645e-07 & 3.5284e-07 \\ -2.1998e-04 & 1.4754e-04 & 1.1228e-04 & 1.9762e-07 & -7.0126e-08 & 1.8670e-07 \\ -8.4608e-07 & 4.9912e-07 & 1.9762e-07 & 9.1702e-10 & -2.6319e-10 & 5.8781e-10 \\ 2.8907e-07 & -1.4645e-07 & -7.0126e-08 & -2.6319e-10 & 1.3575e-10 & -1.9947e-10 \\ -5.8198e-07 & 3.5284e-07 & 1.8670e-07 & 5.8781e-10 & -1.9947e-10 & 4.4082e-10 \end{bmatrix}$$

Table 8: Navigation Results - J2 Perturbation Included

Troll-Svalbard are evaluated and compared. The procedure presented in Section 2.3 is again implemented for each combination. Hence the navigation problem is solved , and the standard deviations of the semi-major axis σ_a and inclination σ_i are computed.

Results Table 9 presents the results for the three station combinations, together with the associated costs. Among the combinations, Kourou-Svalbard achieves the best precision for both orbital parameters, thanks to its geometric diversity. The high latitude of the Svalbard station enhance the precision of the inclination. In tandem, the quasi-equatorial position of Kourou improves the measurement of the SMA. Kourou-Troll performs slightly worse due to reduced latitude and longitude diversity, while Troll-Svalbard achieves accurate estimates for inclination but poorer results for the semi-major axis, since both stations are located at high latitudes. Across the provided time interval, the Kourou-Svalbard combination is the optimal choice to solve the navigation problem, providing the best precision within the budget constraints.

Station Combination	Cost (€)	σ_{SMA} [km]	σ_i [deg]
Kourou-Troll	65,000	1.08946	0.00414
Kourou-Svalbard	65,000	0.00259	0.00032
Troll-Svalbard	70,000	0.03011	0.00052

Table 9: Trade-off analysis: Results for different station combinations.

2.5 Long-Term Analysis

Considering a nominal operational scenario with no time constraints, it is important to prioritize not only measurement accuracy but also reliability and repeatability of data acquisition.

The Kourou-Svalbard combination, which was identified as the most accurate in Section 2.4, might become less desirable in the long term because of Earth's rotation and due to the variation of the RAAN caused by the J2 effect. These factors lead to reduced visibility windows for Kourou over the course of a day, which could also become irregular in the long term. Since Kourou is almost equatorial, it is less suitable for observing a satellite in a sun-synchronous orbit like SMOS. The same reasoning applies to the Kourou-Troll combination.

Despite its lower accuracy, the Troll-Svalbard combination might prove to be the most suitable in a long-term analysis, as both stations are located near the poles, where the satellite passes more frequently. These considerations are confirmed by Fig. 8, which shows the elevation values that guarantee visibility from the three stations over a single day. A potential solution

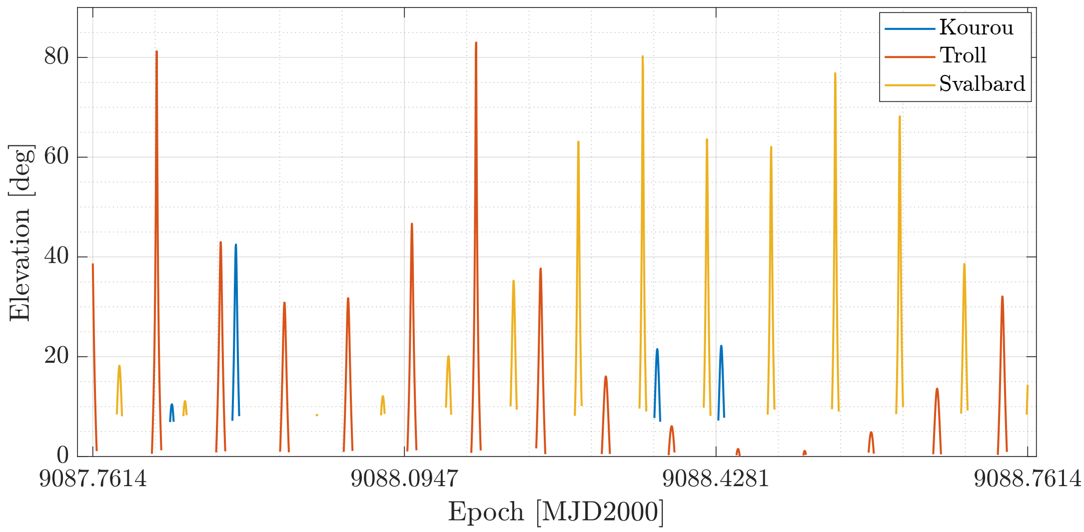


Figure 8: Admissible elevations of the satellite during 1 day.

could be to select Troll and Svalbard as the primary combination and use Kourou as a backup to improve the precision of the semi-major axis measurement.

A compromise between accuracy and reliability could be achieved by introducing additional ground stations, such as those in Malargüe, Perth, and Maspalomas. These stations are located at lower latitudes but are still relatively close to the poles, and their better-spaced longitudes provide improved coverage across the Earth's surface.

Exercise 3: Sequential filters

An increasing number of lunar exploration missions will take place in the next years, many of them aiming at reaching the Moon's surface with landers. In order to ensure efficient navigation performance for these future missions, space agencies have plans to deploy lunar constellations capable of providing positioning measurements for satellites orbiting around the Moon.

Considering a lander on the surface of the Moon, you have been asked to improve the accuracy of the estimate of its latitude and longitude (considering a fixed zero altitude). To perform such operation you can rely on the use of a lunar orbiter, which uses its Inter-Satellite Link (ISL) to acquire range measurements with the lander while orbiting around the Moon. At the same time, assuming the availability of a Lunar Navigation Service, you are also receiving measurements of the lunar orbiter inertial position vector components, such that you can also estimate the spacecraft state within the same state estimation process.
To perform the requested tasks you can refer to the following points.

1. *Check the visibility window.* Considering the initial state $\mathbf{x}_0$ and the time interval with a time-step of 30 seconds from t_0 to t_f reported in Table 10, predict the trajectory of the satellite in an inertial Moon-centered reference frame assuming Keplerian motion. Use the estimated coordinates given in Table 11 to predict the state of the lunar lander. Finally, check that the lander and the orbiter are in relative visibility for the entire time interval.

2. *Simulate measurements.* Always assuming Keplerian motion to model the lunar orbiter dynamics around the Moon, compute the time evolution of its position vector in an inertial Moon-centered reference frame and the time evolution of the relative range between the satellite and the lunar lander. Finally, simulate the measurements by adding a random error to the spacecraft position vector and to the relative range. Assume a Gaussian model to generate the random error, with noise provided in Table 10 for both the relative range and the components of the position vector. Verify (graphically) that the applied noise level is within the desired boundary.

3. *Estimate the lunar orbiter absolute state.* As a first step, you are asked to develop a sequential filter to narrow down the uncertainty on the knowledge of the lunar orbiter absolute state vector. To this aim, you can exploit the measurements of the components of its position vector computed at the previous point. Using an Unscented Kalman Filter (UKF), provide an estimate of the spacecraft state (in terms of mean and covariance) by sequentially processing the acquired measurements in chronological order. To initialize the filter in terms of initial covariance, you can refer to the first six elements of the initial covariance $\mathbf{P}_0$ reported in Table 10. For the initial state, you can perturb the actual initial state $\mathbf{x}_0$ by exploiting the MATLAB function `mvnrnd` and the previously mentioned initial covariance. We suggest to use $\alpha = 0.01$ and $\beta = 2$ for tuning the UT in this case. Plot the time evolution of the error estimate together with the 3σ of the estimated covariance for both position and velocity.

4. *Estimate the lunar lander coordinates.* To fulfill the goal of your mission, you are asked to develop a sequential filter to narrow down the uncertainty on the knowledge of the lunar lander coordinates (considering a fixed zero altitude). To this aim, you can exploit the measurements of the components of the lunar orbiter position vector together with the measurements of the relative range between the orbiter and the lander computed at the previous point. Using an UKF, provide an estimate of the spacecraft state and the lunar lander coordinates (in terms of mean and covariance) by sequentially processing the acquired measurements in chronological order. To initialize the filter in terms of initial covariance, you can refer to the initial covariance $\mathbf{P}_0$ reported in Table 10. For the initial state, you can perturb the actual initial state, composed by $\mathbf{x}_0$ and the latitude

and longitude given in Table 11, by exploiting the MATLAB function `mvnrnd` and the previously mentioned initial covariance. We suggest to use $\alpha = 0.01$ and $\beta = 2$ for tuning the UT in this case. Plot the time evolution of the error estimate together with the 3σ of the estimated covariance for both position and velocity.

Table 10: Initial conditions for the lunar orbiter.

Parameter	Value
Initial state $\mathbf{x}_0$ [km, km/s]	$\mathbf{r}_0$ = [4307.844185282820, -1317.980749248651, 2109.210101634011] $\mathbf{v}_0$ = [-0.110997301537882, -0.509392750828585, 0.815198807994189]
Initial time t_0 [UTC]	2024-11-18T16:30:00.000
Final time t_f [UTC]	2024-11-18T20:30:00.000
Measurements noise	$\sigma_p = 100$ m
Covariance $\mathbf{P}_0$ [km^2, km^2/s^2, rad^2]	`diag([10,1,1,0.001,0.001,0.001,0.00001,0.00001])`

Table 11: Lunar lander - initial guess coordinates and horizon mask

Lander name	MOONLANDER
Coordinates	LAT = 78° LON = 15° ALT = 0 m
Minimum elevation	0 deg

Exercise 3: Sequential filters

3.1 Check the Visibility Window

Methodology The initial state $\mathbf{x}_0$ is propagated within the prescribed time interval, assuming Keplerian motion. The state of the lander is retrieved using the SPICE function `cspice_latrec` to transform the coordinates into the Moon Centered Moon Fixed (MCMF) reference frame and then rotated at each time step into the Moon-Centered Inertial (MCI) reference frame using the function `cspice_sxform`.

A function similar to the one presented in Section 2.1 was used to calculate the elevation, azimuth, and range measurements. In this case, instead of extracting the lander's state from a kernel, the above discussed lander state was directly compared to the orbiter's trajectory. The evolution of the elevation over time is plotted, and it is verified that the elevation remained above the minimum required threshold of 0° throughout the interval.

Results The final state of the satellite is reported in Table 12. Fig. 9 shows that the elevation remains consistently well above the threshold set by the lander, ensuring continuous visibility throughout the time interval.

x [km]	y [km]	z [km]	v_x [km/s]	v_y [km/s]	v_z [km/s]
76.1256	-961.0675	1445.3764	0.0009	0.0002	0.0001

Table 12: Initial state of the lunar lander in the Moon-Centered Inertial (MCI) frame.

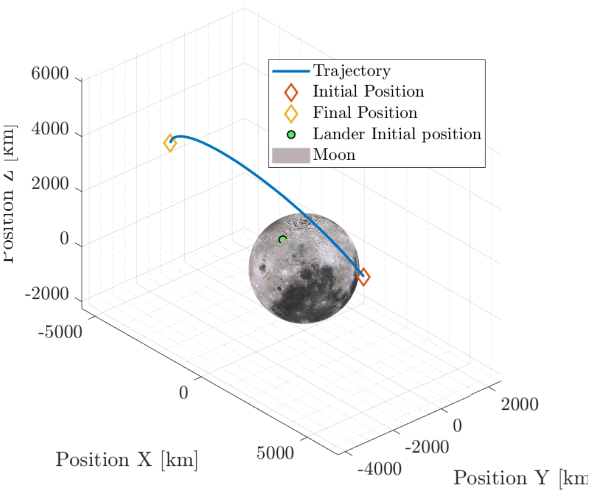


Figure 9: Orbit propagation. (*@Moon MCI*)

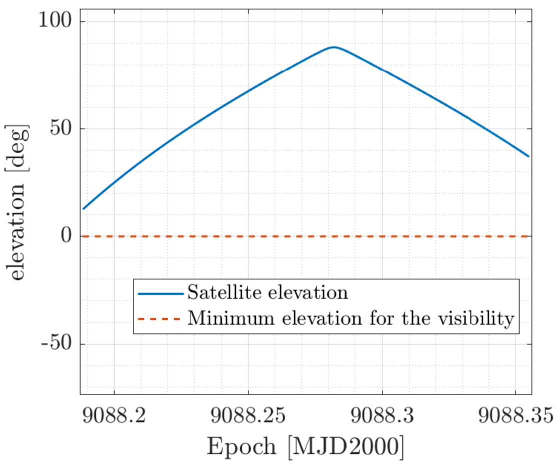


Figure 10: Orbiter elevation with respect to the lander

3.2 Simulate Measurements

Methodology Noise s is added to the position vector using the MATLAB function `mvnrnd`, with the propagated position vector and the square of the related error standard deviations σ_p^2 as input. This time the range is computed with the same procedure as Section 3.1, but retrieving the lander position from the provided kernel in the TOPO frame, then rotating it into the Moon-Centered Inertial (MCI) frame with the SPICE function `cspice_sxform`. The resulting range values are then perturbed using again `mvnrnd` with the variance σ_p^2. Finally, the relative error between the perturbed and unperturbed solutions is plotted over time, along with the associated 3σ confidence level, to assess the Gaussianity of the applied noise.

Results Fig. 11 shows that the relative errors for both position and range fit well within the 3σ distribution, confirming the correct implementation of the noise model.

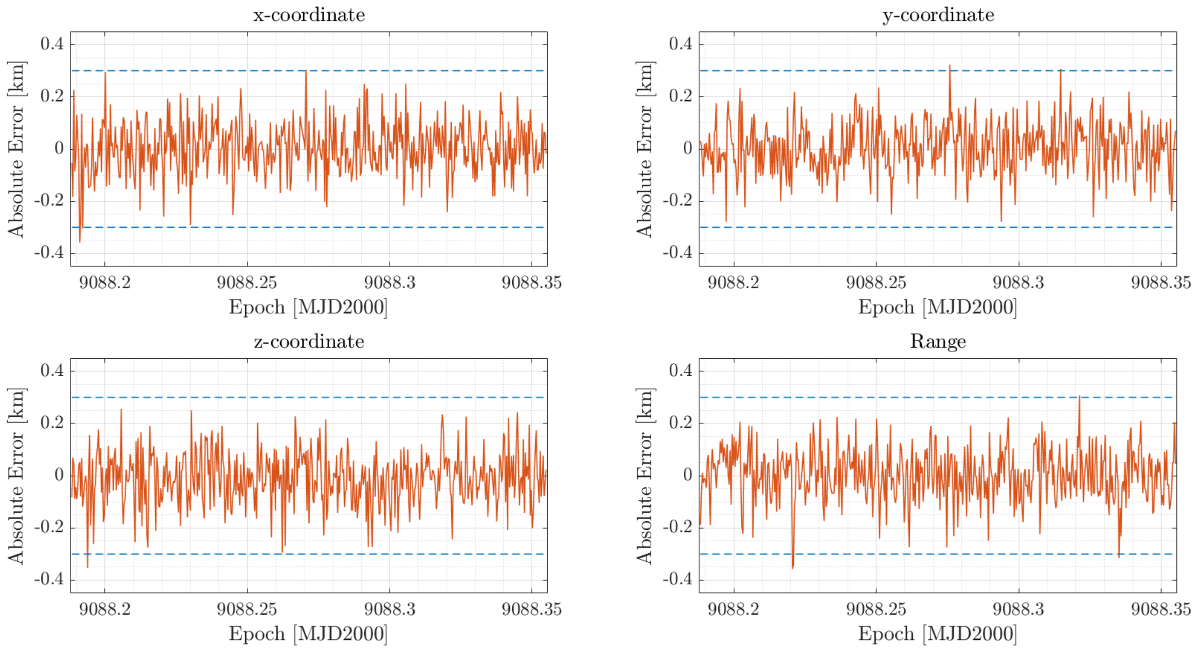


Figure 11: Relative errors for position and range with 3σ confidence level over time.

3.3 Estimate the Lunar Orbiter Absolute State

Methodology To reduce uncertainties during propagation, an Unscented Kalman Filter (UKF) is implemented. The initial condition is generated using the MATLAB function `mvnrnd`, which perturbed the initial state of the satellite with a covariance $\mathbf{P}_{6\times6}$, sub-matrix of the one in Table 10. The weights for the filter were computed using $\alpha = 0.01$ and $\beta = 2$, which were selected for tuning the Unscented Transform (UT).

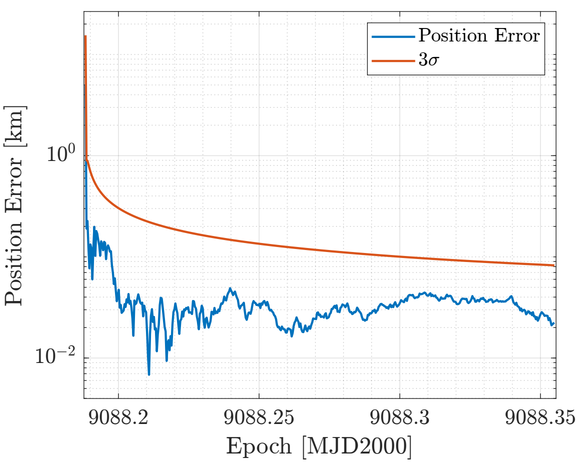
At each step, the propagated state was updated using the available measurements and the computed Kalman gain, providing a refined estimate of the mean and covariance for the satellite's state vector. The flow of the process is summed up in Algorithm 2.

Results The results of the final state and its covariance matrix are reported in Table 13. Comparing these values with the initial covariance matrix $\mathbf{P}_0$, it is clear that the filter significantly improves the precision of the state estimation.

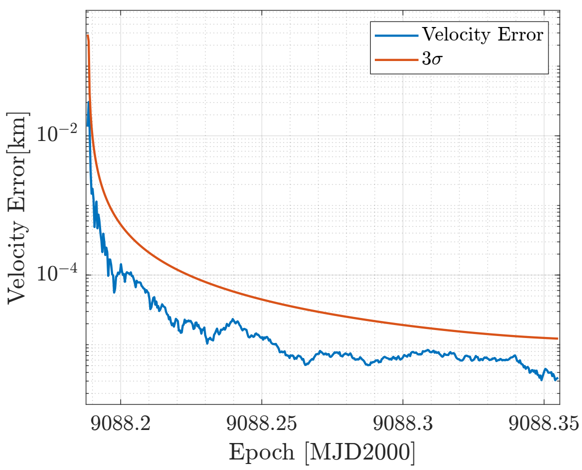
To analyze the evolution of the state estimation over time, Fig. 12 shows the position norm (a) and velocity norm (b) along with their respective 3σ confidence intervals. From the plots, it is evident that the uncertainties decrease rapidly during the initial phase and subsequently stabilize asymptotically, demonstrating the effectiveness of the filter.

Algorithm 2 Unscented Kalman Filter (UKF)

Require: $\mathbf{x}_0$: Initial state, $\mathbf{P}_0$: Initial covariance, Weights: UKF weights, λ: Scaling parameter, t_{span}: Time vector, $\mathbf{y}_{\text{meas}}$: Measurements, $\mathbf{R}$: Measurement noise covariance
Ensure: $\hat{\mathbf{x}}$: State estimates, $\mathbf{P}$: Covariance estimates
  Initialize $\hat{\mathbf{x}}_0 \leftarrow \mathbf{x}_0$, $\mathbf{P}_0 \leftarrow \mathbf{P}_0$
  Pre-allocate solutions $\hat{\mathbf{x}}$ and $\mathbf{P}$ arrays
  Store initial values: $\hat{\mathbf{x}}[1] \leftarrow \mathbf{x}_0$, $\mathbf{P}[1] \leftarrow \mathbf{P}_0$
  for each time step $k = 1, \dots, \text{length}(t_{\text{span}}) - 1$ **do**
    Sigma Point Generation:
      Compute $\sqrt{(n+\lambda)\mathbf{P}_{k-1}}$ with Cholesky decomposition
      Generate sigma points $\boldsymbol{\chi_j}$
    Sigma Point Propagation:
    for each sigma point $\boldsymbol{\chi_j}$ **do**
      Propagate $\boldsymbol{\chi_j}$ using Keplerian motion
    end for
      Compute predicted mean $\hat{\mathbf{x}}_k$ and covariance $\mathbf{P}_k^-$ from the propagated sigma points
    Measurement Prediction:
      Map propagated sigma points to measurement space $\boldsymbol{\gamma_j} = h(\boldsymbol{\chi_j})$
      Compute predicted measurement mean $\hat{\mathbf{y}}_k$
    Update Step:
      Compute covariance $\mathbf{P}_{yy}$ adding the noise $\mathbf{R}$
      Compute cross-covariance $\mathbf{P}_{xy}$
      Calculate Kalman Gain: $\mathbf{K} \leftarrow \mathbf{P}_{xy}\mathbf{P}_{yy}^{-1}$
      Update mean and covariance:
        $\hat{\mathbf{x}}_k^+ = \hat{\mathbf{x}}_k + \mathbf{K}(\mathbf{y}_{\text{meas}} - \hat{\mathbf{y}}_k)$
        $\mathbf{P}_k^+ = \mathbf{P}_k^- - \mathbf{K}\mathbf{P}_{yy}\mathbf{K}^\top$
      Store updated state: $\hat{\mathbf{x}}[k+1] \leftarrow \hat{\mathbf{x}}_k, \mathbf{P}[k+1] \leftarrow \mathbf{P}_k$
      Update variables: $\hat{\mathbf{x}}_{k-1} \leftarrow \hat{\mathbf{x}}_k, \mathbf{P}_{k-1} \leftarrow \mathbf{P}_k$
  end for
    return $\hat{\mathbf{x}}, \mathbf{P}$



(a) Position norm and 3σ confidence interval over time.



(b) Velocity norm and 3σ confidence interval over time.

Figure 12: Time evolution of position and velocity norms with 3σ confidence intervals.

Parameter	Value
Position in MCI $[x, y, z]$ [km]	$[-3479.0345, -2506.2039, 4010.7463]$
Velocity in MCI $[v_x, v_y, v_z]$ [km/s]	$[-0.4501, 0.3485, -0.5578]$

$$P_f = \begin{bmatrix} 7.0297e-05 & 6.7190e-06 & -1.0763e-05 & 5.6061e-09 & 3.4828e-09 & -5.5724e-09 \\ 6.7190e-06 & 8.0786e-05 & -2.7101e-05 & 4.7967e-09 & 8.6570e-09 & -9.0859e-09 \\ -1.0763e-05 & -2.7101e-05 & 1.0722e-04 & -7.6733e-09 & -9.0852e-09 & 1.7520e-08 \\ 5.6061e-09 & 4.7967e-09 & -7.6733e-09 & 1.6500e-12 & 1.0763e-12 & -1.7223e-12 \\ 3.4828e-09 & 8.6570e-09 & -9.0852e-09 & 1.0763e-12 & 1.8881e-12 & -1.7598e-12 \\ -5.5724e-09 & -9.0859e-09 & 1.7520e-08 & -1.7223e-12 & -1.7598e-12 & 3.6048e-12 \end{bmatrix}$$

Table 13: Final state estimate and final covariance at t_f

3.4 Estimate the Lunar Lander Coordinates

Methodology The goal is to extend the UKF developed in Section 3.3 to estimate simultaneously the orbiter's state and the coordinates of the lunar lander. The estimation process follows the same outline presented in Algorithm 2. A new augmented state was defined, including the latitude and longitude of the lander. Similarly, the input measurements were augmented with the range measurements obtained in Section 3.2.

Small changes to the process flow were introduced to adapt the filter to the new state. The most significant involves the transformation of the propagated sigma points into the measurement space. This transformation is no longer trivial, as the lander's latitude and longitude are not directly measured. To compute the range, the same procedure outlined in Section 3.2 was adopted.

Results The final state and covariance of the orbiter are reported in Table 14 and Table 15. Comparing these values with the results obtained in Section 3.3, it can be noticed that the state estimates and its associated uncertainties are slightly more constrained, thanks to the addition of a new measurement.

As in Section 3.3, the evolution of the position and velocity norm absolute errors is plotted alongside the previously obtained results. Fig. 13 shows that the standard deviation stabilizes earlier and at slightly lower values, while the absolute error on the mean does not exhibit significant improvements.

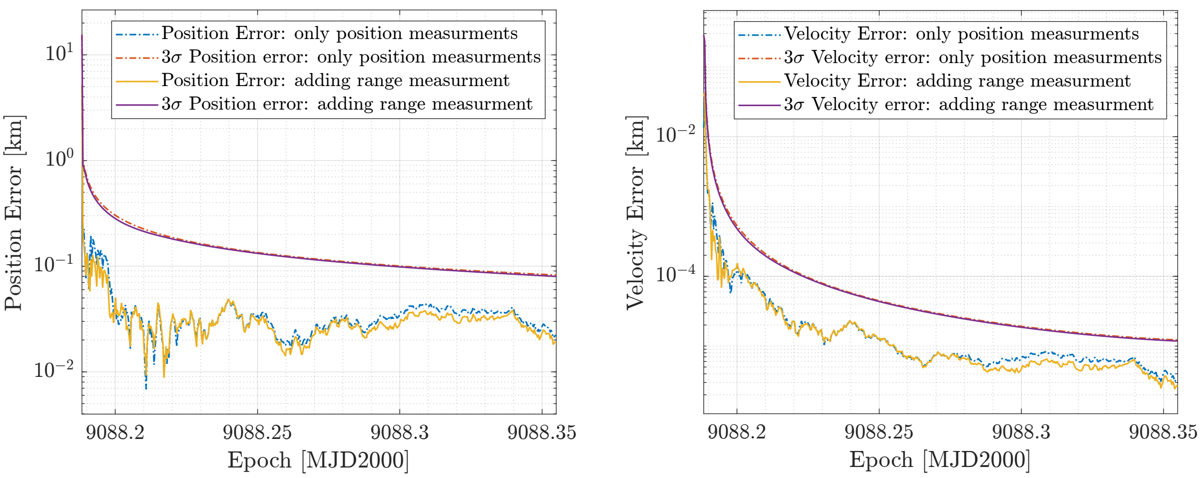
Finally, the estimation of the lander's coordinates, latitude and longitude, is presented in Table 14. Fig. 14 reveals that that the errors between estimated and kernel's values is not always within the 3σ confidence level. This behaviour could be attributed to non-linearities in the relationship between orbiter position and relative range measurements.

Parameter	Value
Orbiter Position in MCI $[x, y, z]$ [km]	$[-3479.0303, -2506.2030, 4010.7450]$
Orbiter Velocity in MCI $[v_x, v_y, v_z]$ [km/s]	$[-0.4501, 0.3485, -0.5578]$
Lander Latitude [deg]	78.237575
Lander Longitude [deg]	15.420411

Table 14: Final state of the lunar orbiter with lander estimation.

$$
\mathbf{P_f} = \begin{bmatrix}
5.9153e-05 & 5.6081e-06 & -8.9774e-06 & 4.2639e-09 & 2.8791e-09 & -4.6039e-09 & -2.5591e-07 & -2.4520e-07 \\
5.6081e-06 & 7.9991e-05 & -2.5872e-05 & 4.4158e-09 & 8.4410e-09 & -8.7381e-09 & -2.6046e-07 & -2.9247e-07 \\
-8.9774e-06 & -2.5872e-05 & 1.0532e-04 & -7.0847e-09 & -8.7491e-09 & 1.6978e-08 & 4.1956e-07 & 4.7290e-07 \\
4.2639e-09 & 4.4158e-09 & -7.0847e-09 & 1.3841e-12 & 9.4515e-13 & -1.5117e-12 & -6.5230e-11 & -5.8407e-11 \\
2.8791e-09 & 8.4410e-09 & -8.7491e-09 & 9.4515e-13 & 1.8187e-12 & -1.6482e-12 & -4.7253e-11 & -5.4952e-11 \\
-4.6039e-09 & -8.7381e-09 & 1.6978e-08 & -1.5117e-12 & -1.6482e-12 & 3.4252e-12 & 7.4619e-11 & 8.7339e-11 \\
-2.5591e-07 & -2.6046e-07 & 4.1956e-07 & -6.5230e-11 & -4.7253e-11 & 7.4619e-11 & 1.3263e-08 & 1.1174e-08 \\
-2.4520e-07 & -2.9247e-07 & 4.7290e-07 & -5.8407e-11 & -5.4952e-11 & 8.7339e-11 & 1.1174e-08 & 1.0144e-08
\end{bmatrix}
$$

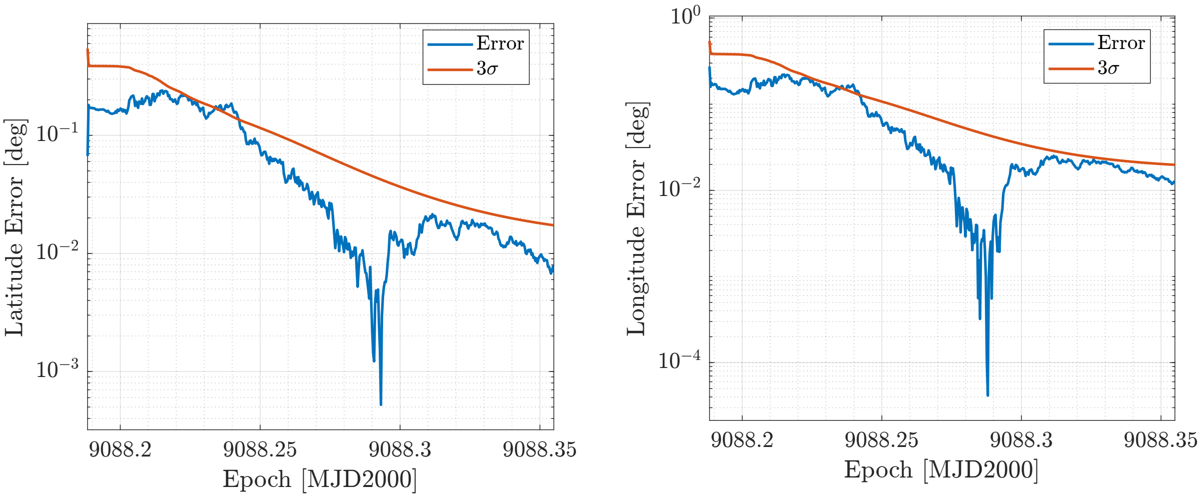
Table 15: Final covariance matrix of the lunar orbiter with lander estimation at t_f.



(a) Comparison of position norm errors with 3σ confidence intervals.

(b) Comparison of velocity norm errors with 3σ confidence intervals.

Figure 13: Comparison of the position (a) and velocity (b) norm errors between Section 3 and the current analysis.



(a) Latitude error with 3σ confidence level.

(b) Longitude error with 3σ confidence level.

Figure 14: Comparison of latitude (a) and longitude (b) errors with respect to kernel.